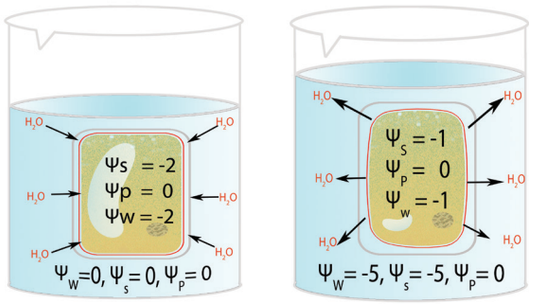



HYPOTONIC SOLUTION HYPERTONIC SOLUTION

**Figure 11.7:** Types of solution based on concentration

### 1. Types of osmosis

Based on the direction of movement of water or solvent in an osmotic system, two types of osmosis can occur, they are ***Endosmosis*** and ***Exosmosis***.

***i.*** ***Endosmosis***: Endosmosis is defined as the osmotic entry of solvent into a cell or a system when it is placed in a pure water or hypotonic solution.
For example, dry raisins (high solute and low solvent) placed in the water, it swells up due to turgidity.

***ii.*** ***Exosmosis***: Exosmosis is defined as the osmotic withdrawal of water from a cell or system when it is placed in a hypertonic solution. Exosmosis in a plant cell leads to **plasmolysis**.

### 2. Plasmolysis (*Plasma* = cytoplasm; *lysis* = breakdown)

When a plant cell is kept in a hypertonic solution, water leaves the cell due to **exosmosis**. As a result of water loss, protoplasm shrinks and the cell membrane is pulled away from the cell wall and finally, the cell becomes **flaccid**. This process is named as **plasmolysis**.

Wilting of plants noticed under the condition of water scarcity is an indication of plasmolysis. Three types of plasmolysis occur in plants: i) ***Incipient plasmolysis ii) Evident plasmolysis*** **and** ***iii) Final plasmolysis***. Differences among them are given in table 11.2.

### Significance

Plasmolysis is exhibited only by living cells and so it is used to test whether the cell is living or dead.

### 3. Deplasmolysis

The effect of plasmolysis can be reversed, by transferring them back into water or **hypotonic solution**. Due to endosmosis, the cell becomes turgid again. It regains its original shape and size. This phenomenon of the revival of the plasmolysed cell is called **deplasmolysis**. Example: Immersion of dry raisin in water.

#### Potato Osmoscope

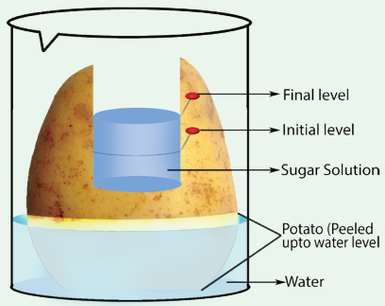



**Figure 11.8:** Demonstration of Endosmosis by Potato Osmoscope

i. Take a peeled potato tuber and make a cavity inside with the help of a knife.
ii. Fill the cavity with concentrated sugar solution and mark the initial level.
iii. Place this setup in a beaker of pure water.
iv. After 10 minutes observe the sugar solution level and record your findings (Figure 11.8).
v. With the help of your teacher discuss the results.

Instead of potato use beetroot or bottle-guard and repeat the above experiment. Compare and discuss the results.

### 4. Reverse Osmosis

Reverse Osmosis follows the same principles of osmosis, but in the reverse direction. In this process movement of water is reversed by applying pressure to force the water against a concentration gradient of the solution. In regular osmosis, the water molecules move from the higher concentration (pure water = hypotonic) to lower concentration (salt water = hypertonic). But in reverse osmosis, the water molecules move from the lower concentration (salt water = hypertonic) to higher concentration (pure water = hypotonic) through a selectively permeable membrane (Figure 11.9).



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